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	MICRO LABS LIMITED API, BOMMASANDRA (ML-15)	Page 1 of 2
		No: RMSPIH175
	SPECIFICATION	Revision No:04
PRODUCT	ETHYLENE DICHLORIDE	Effective Date: 29/03/22
		Next Review : Feb 2027

1.0 PRODUCT INFORMATION

CATEGORY	Raw Material
FORMULA	Chemical Name : 1,2-Dichloroethane Alternative name : Ethylene dichloride Molecular Weight : 98.96 g/mol Molecular Formula : $C_2H_4Cl_2$ CAS No. : 107-06-2 Product code : EDC Item Code : 100S0200D027
SAMPLING PROCEDURE	Sampling to be performed as per current version of SOP No. SOP:QCGN:008
TESTING PROCEDURE	Testing to be performed as per current version of STP No. RMTPIH175
STORAGE CONDITIONS	Store in well closed containers below 35°C
HANDLING PRECAUTIONS (if any)	Use goggles, hand gloves and nose mask while sampling and handling the material.
PACKAGING CONDITION	Pack the sample in air tight container protected from light.
SAMPLING QUANTITY	50 mL

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2.0 SPECIFICATION

S. No.	TEST	SPECIFICATION LIMIT
1.	Description	Clear colorless liquid.
2.	Identification i) By IR ii) By GC	 The IR spectrum of the sample should be concordant with the IR spectrum of the Ethylene dichloride reference/working standard. The retention time of the major peak in the chromatogram of the sample solution corresponds to that in the chromatogram of the reference/working standard solution as obtained in the test for Related substances by GC.
3.	Miscibility	Miscible in methanol and acetone.
4.	Water content by KF titrator	Not more than 0.3% w/w
5.	Residue on evaporation	Not more than 0.1% w/w
6.	Related Substance by GC • Any unspecified impurity • Total impurities	 Not more than 0.15 % Not more than 0.3 %
7.	Assay by GC (on anhydrous basis)	Not less than 99.0% w/w

Chemical Names: Not applicable

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1. Description:

Transfer about 10 mL of sample into a clean and dry test tube; observe the color, clarity and presence of extraneous matter under a white background.

2. Identification

i) By IR:

Standard preparation:

Put 1 to 2 drops of the neat standard on a KBr or NaCl plates and carefully place another plate on top of this taking care not to form any air bubble between the two plates. Place the plates in a holder and expose to the beam of the IR Spectrophotometer. Record the spectrum of the standard preparation over the range from about 4000 cm^{-1} to 650 cm^{-1}. Perform the back ground correction using the plates without the standard and analyzing over the same range as that of the standard before obtaining the spectrum of the standard.

Sample preparation:

Put 1 to 2 drops of the neat sample on a KBr or NaCl plates and carefully place another plate on top of this taking care not to form any air bubble between the two plates. Place the plates in a holder and expose to the beam of the IR Spectrophotometer. Record the spectrum of the test preparation over the range from about 4000 cm^{-1} to 650 cm^{-1}.

Interpretation: The IR spectrum of the sample should be concordant with the IR spectrum of the Ethylene dichloride reference/working standard.

ii) By GC:

The test is perform as per the procedure given in section 6 for Related substances by GC.

Interpretation: The retention time of the major peak in the chromatogram of the sample solution corresponds to that in the chromatogram of the reference/working standard solution as obtained in the test for Related substances by GC.

3. Miscibility:

Transfer 10 mL of sample into a 100 mL stopper test tube and add 10 mL of methanol. Shake for 2 minutes. Observe the miscibility and record the observation.

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Transfer 10 mL of sample into a 100 mL stopper test tube and add 10 mL of acetone. Shake for 2 minutes. Observe the miscibility and record the observation.

4. **Water content by KF titrator:**

Transfer about 50 mL of Methanol into the titration vessel and neutralize the methanol using Karl Fischer reagent .After the end point has reached, weigh about 2.0 g of the sample transfer quickly into the titration vessel. Use a stirrer to dissolve the sample and then titrate it to the end point with Karl Fisher reagent.

Perform the water content analysis in duplicate and report the average result.

Calculation:

$$\text{Water content (\%w/w)} = \frac{\text{Volume of KFR consumed x KFR factor x 100}}{\text{Weight of the sample (in g) x 1000}}$$

5. **Residue on evaporation:**

Weigh accurately a porcelain dish that has been previously dried at 105°C for about 30 minutes and cooled to room temperature in desiccator (W_1).Accurately weigh about 10 g of sample into a porcelain dish and record the weight (W_2) and heat the porcelain dish along with sample on a hot plate at about 150°C to 200°C in a fume hood and evaporate to dryness. Then place the porcelain dish along with residue in oven and dry for about 30 minutes at 105° C and cool to room temperature in desiccators and weigh the porcelain dish (W_3).

$$\text{Residue on evaporation (\% w/w)} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \times 100$$

6. **Related Substances by GC:**

Chromatographic condition: A gas chromatograph equipped with FID and integrator.

Column	:	DB-5 (30m, 0.53mm ID, 5.0µm)
Column catalog No.	:	125-5035
Injector temperature	:	180° C
Detector temperature	:	250° C
Split ratio	:	1:40

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Column flow : 3.0 mL/minute

Injection volume : 0.2 µL

Retention time : The retention time of the Ethylene dichloride is about 10.6 minutes.

Oven temperature : Initially hold at 40 °C for 5 minutes, then increase the temperature to 180 °C at a rate of 10 °C/minute and hold at 180°C for 0 minutes. Finally, increase the temperature to 250 °C at 20°C/minutes and hold at 250 °C for 5minutes.

Standard preparation: Inject the Ethylene dichloride standard as such.

Sample solution: Inject the sample as such

Procedure:

Set up the gas chromatogram with the conditions described above and allow the system and the column to equilibrate. Run the system as per the injection sequences given below. Record the chromatograms and measure the peak responses.

Injection sequence table:

Name the solution	No.of injection
System Blank	1
Standard solution	1
Sample solution	1
Standard solution- Bracketing*	1

*Inject intermittent standard solution for every three batches

System suitability criteria:

1. The tailing factor of Ethylene dichloride should not be more than 2.0 for the Standard solution and bracketing Standard solution.
2. The theoretical plates of Ethylene dichloride should not be less 40000 for the Standard solution and bracketing Standard solution.

Reporting of results: Report the result by % area normalization.

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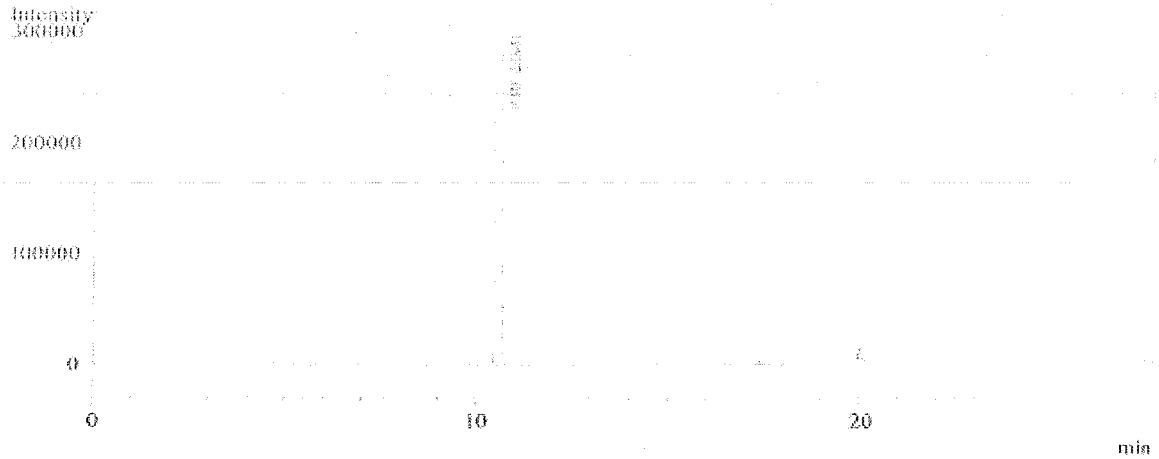
Figure-1 Typical chromatogram of the standard solution in the Related substances by GC



Peak#	Name	Ret.Time	Area	Area%	Theoretical Plates	Tailing Factor
1	EHA-S1	10.59	48411035	100.000	84888	1.07
Total			48411035	100.000		

7. **Assay by GC (On anhydrous basis):**

Chromatographic condition: A gas chromatograph equipped with FID and integrator.

Column	:	DB-624 (60 m, 0.32 mm ID, 1.8 µm)
Column catalog No.	:	123-1364
Injector temperature	:	220° C
Detector temperature	:	260° C
Split ratio	:	1:10
Column flow	:	1.5 mL/minute
Injection volume	:	1.0 µL
Retention time	:	The retention time of the Ethylene dichloride is about 12.0 minutes.
Oven temperature	:	Initially hold at 40° C for 5 minutes and then increase the temperature to 240° C at 20° C/minutes and hold at 240° C for 5 minutes.
Diluent	:	Methanol.

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Standard solution preparation: Accurately weigh about 250 mg of Ethylene dichloride reference/working standard into a 25 mL volumetric flask containing 10 mL of diluent, dilute and make up to the mark with diluent. Shake well to mix.

Sample solution preparation: Accurately weigh about 250 mg of sample into a 25 mL volumetric flask containing 10 mL of diluent, dilute and make up to the mark with diluent. Shake well to mix.

Procedure:

Set up the gas chromatogram with the conditions described above and allow the system and the column to equilibrate. Run the system as per the injection sequences given below. Record the chromatograms and measure the peak responses.

Injection sequence table:

Name the solution	No. of injection
System Blank	1
Blank	1
Standard solution	5
Blank	1
Sample solution	2
Standard solution- Bracketing*	1

*Inject intermittent standard solution for every three batches

System suitability criteria:

1. The tailing factor of Ethylene dichloride peak should not be more than 2.0 obtained from the first standard solution and bracketing standard solution.
2. % RSD of area of Ethylene dichloride peak obtained from five replicate injections of standard solution and bracketing standard solution should not be more than 5.0

Calculation:

Assay (on anhydrous basis, % w/w) =

$$\frac{\text{Average corrected area of Ethylene dichloride peak in sample solution}}{\text{Average corrected area of Ethylene dichloride from 5 replicate standard injections}} \times \frac{\text{Wt. of the standard}}{\text{Wt. of the sample}} \times \frac{100}{(100\text{- water content})} \times P^*$$

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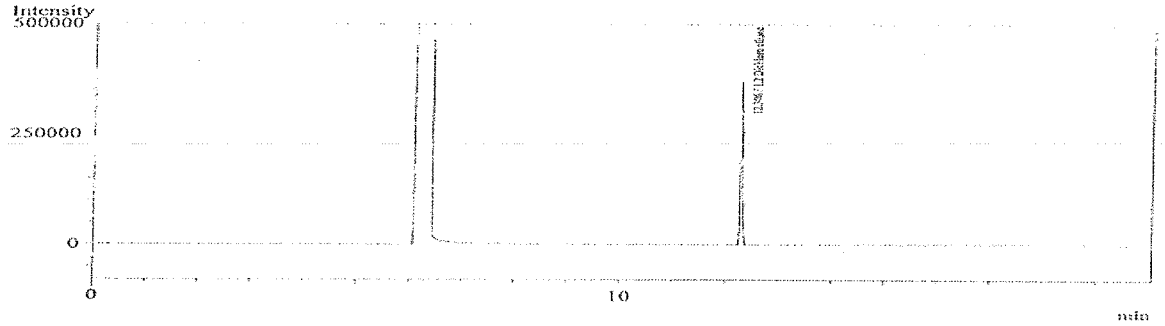
P*: Potency of the Ethylene dichloride reference/working standard.

Figure-2: Typical chromatogram of the standard solution in assay by GC.



Peak#	Name	Ret.Time	Area	Area%	Tailing Factor
1	1,2 Dichloro ethane	12.31	1035831	100.00	1.083
Total			1035831	100.00	

REVISION SUMMARY:

Revision Number	CCF Number	Effective Date.	Revision History
00	CCF14062	19/03/2014	New
01	CCF14295	06/11/2014	1. Typographic error corrected in Relates substances by GC oven temperature (i.e. 180°C instead of 108°C) 2. Bracketing standard has been incorporated.
02	CCF15322	22/08/2015	1. The following test have been incorporated in Specification and STP ➢ Residue on evaporation ➢ Assay by GC 2. IR scanning range modified from "4000 cm^{-1} to 400 cm^{-1}" to "4000 cm^{-1} to 650 cm^{-1}" in STP. In line with all pharmacopeia. 3. Product title has been modified from

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Revision Number	CCF Number	Effective Date.	Revision History
			"1,2-Dichloroethane (EHA-S1)" to "EHA-S1" in Specification and STP. 4. Editorial changes have been done in Specification and STP for more clarity. 5. Specification and STP formats have been modified in line with current version of SOP QAP:014
03	CCF17270	07/06/2017	1. Specification and STP formats have been modified. To in line with current SOP (SOP:QAP:014) revision. 2. Editorial changes have been done in the Specification and STP. 3. Product title name has been changed from EHA-S1 to Ethylene dichloride in the Specification and STP. 4. Item code has been incorporated in the Specification. 5. Category has been changed from Starting material to Raw material. Accordingly, reserve sample quantity and total quantity of sample have been excluded from Specification. 6. Product code has been excluded from Specification and STP.

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Revision Number	CCF Number	Effective Date.	Revision History
04	CCP-ML15-22-0167	29/03/22	1. As part of periodical review Specification and STP has been revised. 2. Specification and STP formats have been modified. To in line with QAP:014/07. 3. Product code has been incorporated in the specification. 4. The following changes have been made in the Related substances by GC test in Specification and STP. ➢ The terminology "Any other individual impurity" have been renamed as "Any unspecified impurity" in Specification. ➢ Column catalog number has been incorporated for Related substances by GC in STP. ➢ Standard solution injection renamed as Bracketing standard solution after sample solution injection in the injection sequence table. Accordingly, changes are made in the system suitability criteria wherever applicable. 5. The following changes have been made in the Assay by GC test in STP. ➢ Column catalog number has been incorporated for Assay by GC in STP. ➢ Standard solution injection renamed as

Revision Number	CCF Number	Effective Date.	Revision History
			Bracketing standard solution after sample solution injection in the injection sequence table. Accordingly, changes are made in the system suitability criteria wherever applicable.

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